

SkyInsight

Airline Passenger Satisfaction Analytics

Data Visualization & Analytics - Capstone 2 Final Project Report

Detailed Academic + Industry Report

Project Title	SkyInsight - Airline Passenger Satisfaction Analytics
Sector	Aviation / Airline Industry
Team ID	DVA-D-G15, Section D
Institute	Newton School of Technology
Faculty Mentor	Archit Raj
Team Members and Roles	Yogesh Mishra - Project/Data/ETL Lead; Mohammad Affan Anas - Analysis & Visualization Lead; Aditi Singh - Strategy Lead; Bhavya Punj - PPT & Quality Lead
GitHub Repository	https://github.com/yogeshm01/SectionD_G15_SkyInsight
Tableau Public Dashboard	https://public.tableau.com/app/profile/bhavya.punj2072/viz/AirlineAnalysisDashboard1/CustomerExperienceOverview?publish=yes
Submission Date	29 April 2026

Prepared for decision-makers who need a clear view of passenger satisfaction, recommendation behaviour, and priority areas for customer experience improvement.

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1. Executive Summary

This project studies airline passenger reviews to answer a practical business question: what does the customer experience look like across airlines, seat classes, and traveller segments, and where should airlines focus first to improve satisfaction and recommendations? The cleaned analysis dataset contains 22,329 passenger reviews across 497 airlines. The final dataset supplied for this report has six business-ready fields: airline name, overall rating, seat type, traveller type, recommendation flag, and rating category.

The main story is clear: customer dissatisfaction is not a small edge case. The average overall rating is only 3.14 on the available rating scale, the median is 1, and 68.3% of reviews fall into the Low category. Only 34.9% of passengers recommend the airline. This means the project is not only about dashboard reporting; it is about helping airline stakeholders choose the few service-improvement areas most likely to move customer trust.

The analysis uses a Python-to-Tableau workflow. Python was used for extraction checks, cleaning, standardisation, missing-value handling, KPI calculation, segmentation, and statistical testing. Tableau Public was used as the executive-facing layer so that non-technical users can filter by airline, seat type, and traveller type and quickly identify performance gaps.

Executive KPI	Value	Business meaning
Total cleaned reviews	22,329	Large enough to identify broad passenger experience patterns.
Airlines covered	497	Allows benchmarking across a wide competitive set.
Average overall rating	3.14	Overall satisfaction is weak and needs improvement.
Median rating	1	At least half the reviews are very low rated.
Recommendation rate	34.9%	Only about one in three reviewers would recommend.
Low rating share	68.3%	Most reviews require recovery attention.

The highest-impact recommendations are to prioritise Economy and Premium Economy experience recovery, fix pain points for Family and Couple Leisure travellers, benchmark practices from better-rated airlines, and strengthen the ETL pipeline so future dashboards retain service-level fields such as seat comfort, food quality, ground service, and value for money.

2. Sector & Business Context

The aviation sector is highly sensitive to customer experience because passengers often compare airlines on reliability, comfort, staff behaviour, value for money, and the confidence that the airline will handle disruptions fairly. In this environment, a low review score does not only represent one unhappy customer; it can influence future bookings, online reputation, and loyalty programme performance.

The primary decision-maker for this project is an airline customer experience leader, supported by operations, cabin service, digital experience, and revenue teams. The dashboard can also help a strategy or benchmarking team compare airline-level performance and identify where a competitor appears to be doing better.

This problem was chosen because airline review data is easy to understand but still business-critical. Ratings and recommendation behaviour act as a direct voice-of-customer signal. When analysed well, they show where operational investment should go first instead of relying only on broad assumptions.

The business value is practical: the analysis turns thousands of reviews into a smaller set of decision points. Instead of treating customer feedback as scattered text and ratings, the project creates a KPI framework that separates customer segments, seat classes, and airline performance levels.

3. Problem Statement & Objectives

Formal problem definition: Airlines need a clear, data-backed way to understand which customer segments and travel classes are experiencing the weakest satisfaction and how those satisfaction levels connect with passenger recommendation behaviour.

Project objectives

1. Build a clean and dashboard-ready dataset from airline review data.
2. Measure overall satisfaction and passenger recommendation behaviour using simple KPIs.
3. Compare satisfaction across airlines, seat types, and traveller types.
4. Identify risk segments where low satisfaction can damage loyalty and word-of-mouth.
5. Create a Tableau dashboard that supports fast executive review and operational drill-down.

Scope

In scope	Out of scope
Overall rating analysis, recommendation rate, rating category distribution, seat-type and traveller-type segmentation, airline benchmarking, basic statistical testing.	Causal claims, live operational data integration, actual revenue modelling, route-level conclusions, and text sentiment modelling from review comments because the uploaded final CSV does not include review text.
Python cleaning and KPI calculation, Tableau dashboard design, business recommendations, and impact estimation.	Real-time forecasting or machine learning deployment. These are proposed as future scope.

Success criteria

- The dataset is cleaned, standardised, and free of missing values in the final analysis fields.
- KPIs are easy to explain to a business stakeholder.
- The dashboard supports filtering and comparison without requiring Python knowledge.

- Insights lead directly to prioritised actions, not just chart descriptions.

4. Data Description

The project source is the Kaggle Airline Reviews dataset as documented in the project repository. The final CSV supplied for this report is a cleaned, reduced analysis table created from that source. The repository describes the source as a CSV dataset with airline review records and 20+ original columns; the uploaded final file used for this report contains the six columns required for the dashboard-level analysis.

Dataset item	Details
Source name	Kaggle - Airline Reviews Dataset
Direct access link	https://www.kaggle.com/datasets/juhibhojani/airline-reviews
Uploaded final file	final_airline_reviews(1).csv
Rows in uploaded final file	22,329
Columns in uploaded final file	6
Airlines represented	497
Rating range observed	1 to 9
Missing values in final fields	0 in all six columns

Column-by-column explanation of final analysis fields

Column	Type	Business meaning
airline_name	Categorical text	Name of the airline being reviewed. Used for benchmarking and comparison.
overall_rating	Numeric	Passenger overall satisfaction score. Used as the primary rating KPI.
seat_type	Categorical text	Cabin class or travel class, such as Economy Class, Business Class, or Premium Economy.
type_of_traveller	Categorical text	Customer segment, such as Solo Leisure, Couple Leisure, Family Leisure, or Business.
is_recommended	Binary numeric	1 means the passenger recommends the airline; 0 means the passenger does not recommend it.
rating_category	Categorical text	Derived label: Low, Medium, or High, based on overall rating.

Data limitations and known biases

- Public review data is self-selected; very satisfied or very dissatisfied passengers are more likely to post reviews than average passengers.
- The uploaded final file does not include review dates, review text, route, aircraft, or individual service-rating columns, so trend analysis and detailed root-cause attribution are limited.
- Some records contain Unknown categories for seat type or traveller type. These are useful for transparency but should not be treated as a real passenger segment.

- Airline-level comparisons are more reliable when a minimum review count is applied. This report uses a 20+ review threshold for top/bottom airline tables.

5. Data Cleaning & ETL Pipeline

The cleaning approach follows the Python workflow shown in the project notebook: load the raw CSV, inspect structure, standardise column names, remove unnecessary columns, handle missing values, convert data types, engineer recommendation and category fields, and export a clean CSV for Tableau. The aim was not to over-process the data, but to make it reliable enough for KPI reporting and segmentation.

ETL step	What was done	Why it matters
Load and inspect	Loaded the airline review CSV using pandas and checked head, shape, columns, info, describe, and missing values.	Prevents blind dashboard building and confirms what the data actually contains.
Column standardisation	Lower-cased column names, stripped spaces, and replaced spaces/special characters with underscores.	Makes code and Tableau field handling easier and less error-prone.
Column selection	Dropped unnecessary raw fields and prepared a final reduced table for dashboard KPIs.	Keeps the dashboard focused on business questions instead of clutter.
Missing-value handling	Filled important categorical values with Unknown and used numeric imputation in the broader cleaning notebook where service columns were available.	Keeps rows usable while preserving data-quality transparency.
Data type conversion	Converted overall_rating to numeric and date fields to datetime in the notebook workflow.	Allows aggregation, filtering, and statistical testing.
Feature engineering	Mapped yes/no recommendation into is_recommended and created rating_category.	Creates business-readable KPIs and segment labels.
Final export	Saved a clean CSV for Tableau and reporting.	Creates a reproducible handoff between Python and Tableau.

Important correction applied in this report: the recommendation KPI should be calculated from a clean binary flag such as is_recommended. Converting a text column like recommended directly with astype(bool) can incorrectly treat both "yes" and "no" strings as True, producing a misleading 100% recommendation rate. The uploaded final dataset already contains the correct binary is_recommended field, and the correct recommendation rate is 34.9%.

Missing values in uploaded final file

Column	Missing values	Treatment
airline_name	0	No action needed in final CSV
overall_rating	0	No action needed in final CSV
seat_type	0	No action needed in final CSV
type_of_traveller	0	No action needed in final CSV
is_recommended	0	No action needed in final CSV
rating_category	0	No action needed in final CSV

Outlier check

Overall ratings are bounded values, so the outlier strategy was conservative. Using the IQR method, $Q1 = 1$, $Q3 = 5$, $IQR = 4$, lower bound = -5, and upper bound = 11. Since the observed ratings range from 1 to 9, the IQR method identifies 0 statistical outliers. No rating records were removed only for being low or high, because extreme ratings are meaningful customer signals.

6. KPI & Metric Framework

The KPI framework is intentionally simple so that a CEO, customer experience head, or operations manager can understand the result without reading Python code. Each KPI has a clear business question attached to it.

KPI	Formula / computation logic	Why it matters	Project objective mapped
Average Overall Rating	<code>mean(overall_rating)</code>	Shows the general satisfaction level.	Measure overall customer satisfaction.
Recommendation Rate (%)	<code>sum(is_recommended) / count(reviews) x 100</code>	Captures whether satisfaction converts into positive word-of-mouth.	Understand loyalty and advocacy risk.
Low Rating Share (%)	<code>count(rating_category = Low) / total reviews x 100</code>	Shows how much of the review base needs service recovery.	Identify dissatisfaction concentration.
Segment Satisfaction Score	Average rating by seat_type or type_of_traveller	Helps teams choose where to act first.	Compare customer segments.
Airline Benchmark Rating	Average rating by airline_name, using minimum review threshold	Identifies best-practice and high-risk airline groups.	Benchmark airline performance.

Metric	Current value
Average Overall Rating	3.14
Median Overall Rating	1
Recommendation Rate	34.9%
Low Rating Share	68.3%
Medium Rating Share	10.6%
High Rating Share	21.1%

7. Exploratory Data Analysis

The EDA focuses on distribution, comparison, and segment differences. Because the uploaded final file does not contain date fields, trend analysis is described as a future-dashboard requirement rather than calculated from this file.

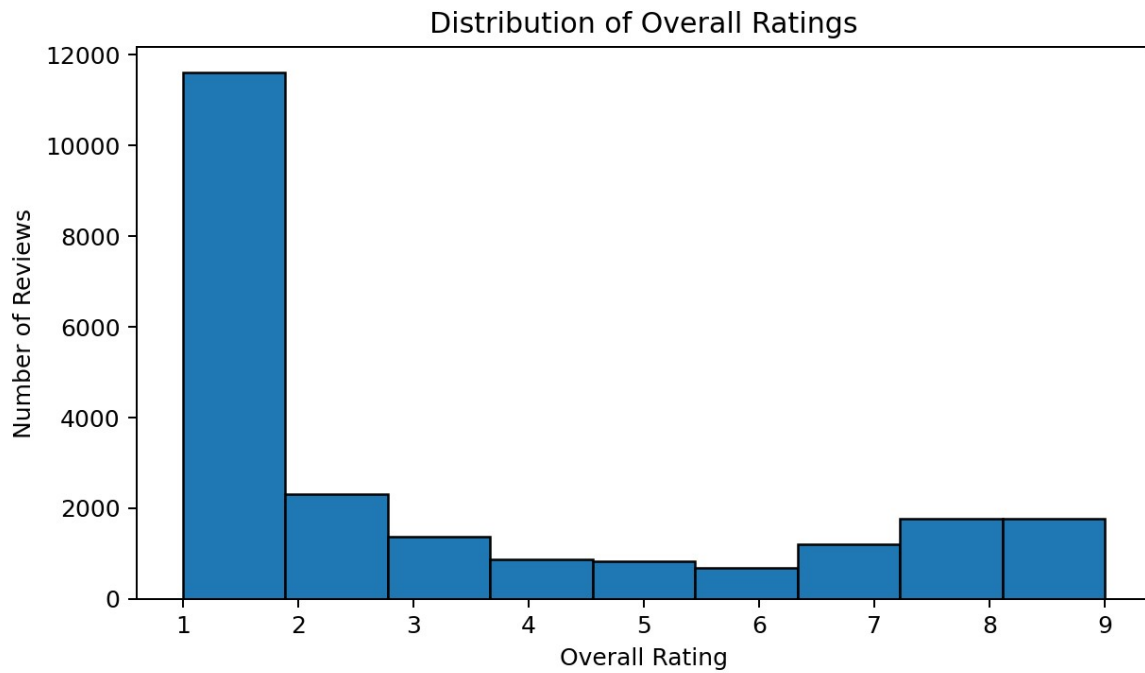


Figure 1. Distribution of overall ratings. The distribution is heavily concentrated at the low end, which explains the weak average and low median.

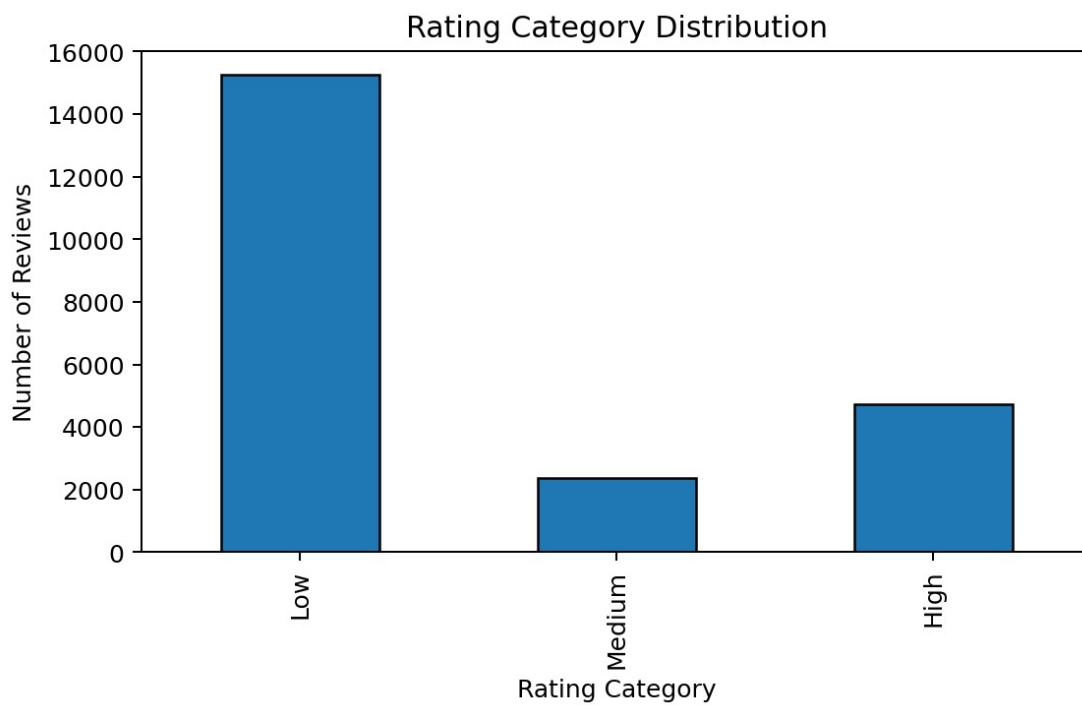


Figure 2. Rating category distribution. Low ratings dominate the review base, making service recovery a core business priority.

Rating category	Reviews	Share	Average rating	Recommendation rate
Low	15,247	68.3%	1.33	15.4%
Medium	2,364	10.6%	4.92	35.5%
High	4,718	21.1%	8.12	97.7%

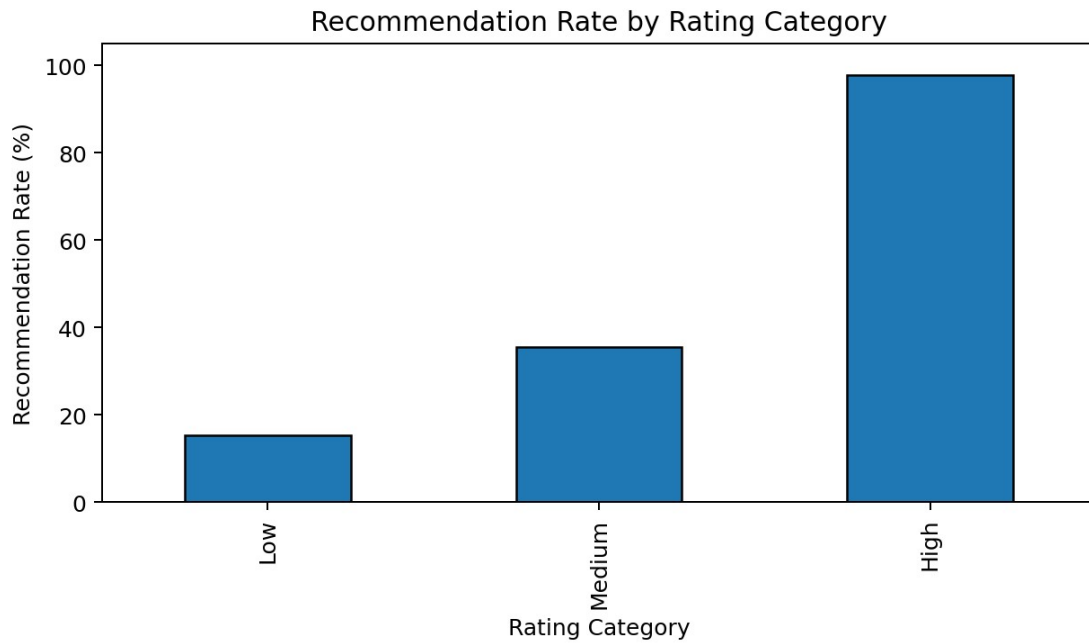


Figure 3. Recommendation rate by rating category. High ratings almost always convert into recommendations, while low ratings rarely do.

Comparison by seat type

Seat type is one of the most useful operational cuts because it maps directly to cabin product, pricing, service design, and passenger expectations. Business Class has the strongest average rating in the final file, while Economy and Premium Economy are weak despite being commercially important.

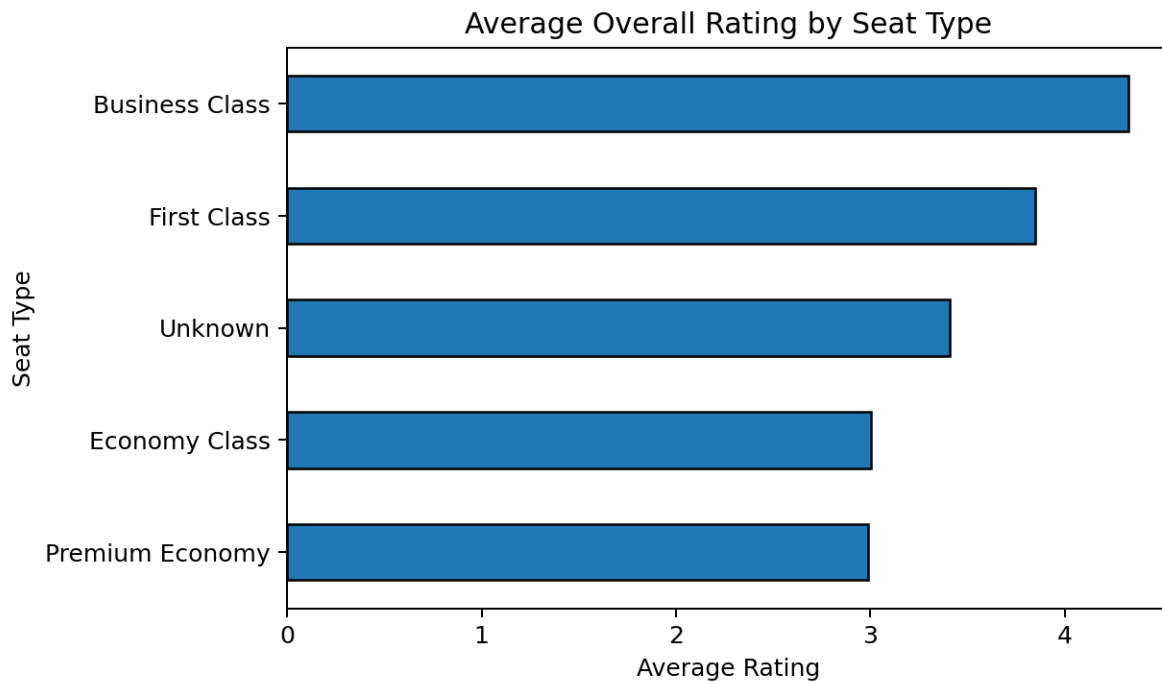


Figure 4. Average overall rating by seat type.

Seat type	Reviews	Average rating	Recommendation rate
Business Class	2,077	4.33	54.8%
First Class	185	3.85	58.4%

Unknown	568	3.41	5.5%
Economy Class	18,853	3.00	33.4%
Premium Economy	646	2.99	34.4%

Comparison by traveller type

Traveller type shows where the airline experience may fail different customer needs. Family Leisure and Couple Leisure passengers show the weakest average ratings, which suggests that disruption handling, seating, baggage, airport experience, and comfort may matter strongly for these groups.

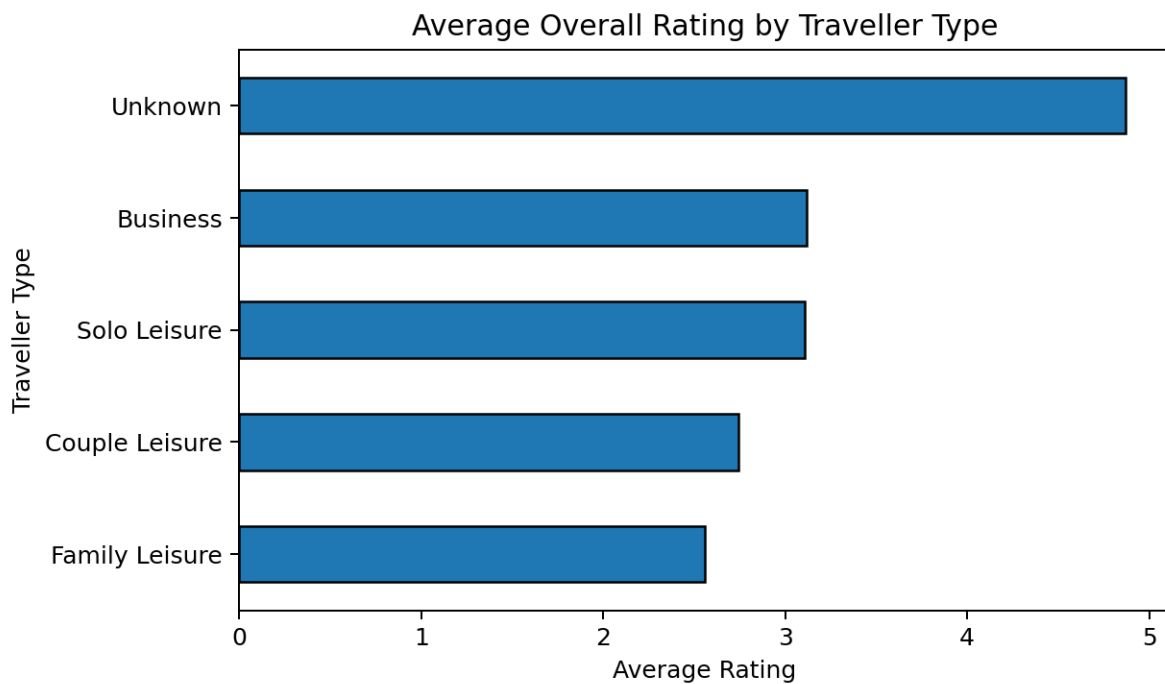


Figure 5. Average overall rating by traveller type.

Traveller type	Reviews	Average rating	Recommendation rate
Unknown	2,896	4.87	58.4%
Business	2,696	3.12	31.6%
Solo Leisure	7,120	3.11	36.5%
Couple Leisure	5,265	2.74	28.4%
Family Leisure	4,352	2.56	26.9%

Airline benchmarking

For airline benchmarking, a minimum threshold of 20 reviews is used to reduce the risk of ranking airlines based on very small samples. The top group gives potential benchmark cases; the bottom group highlights where customer experience risk is severe.

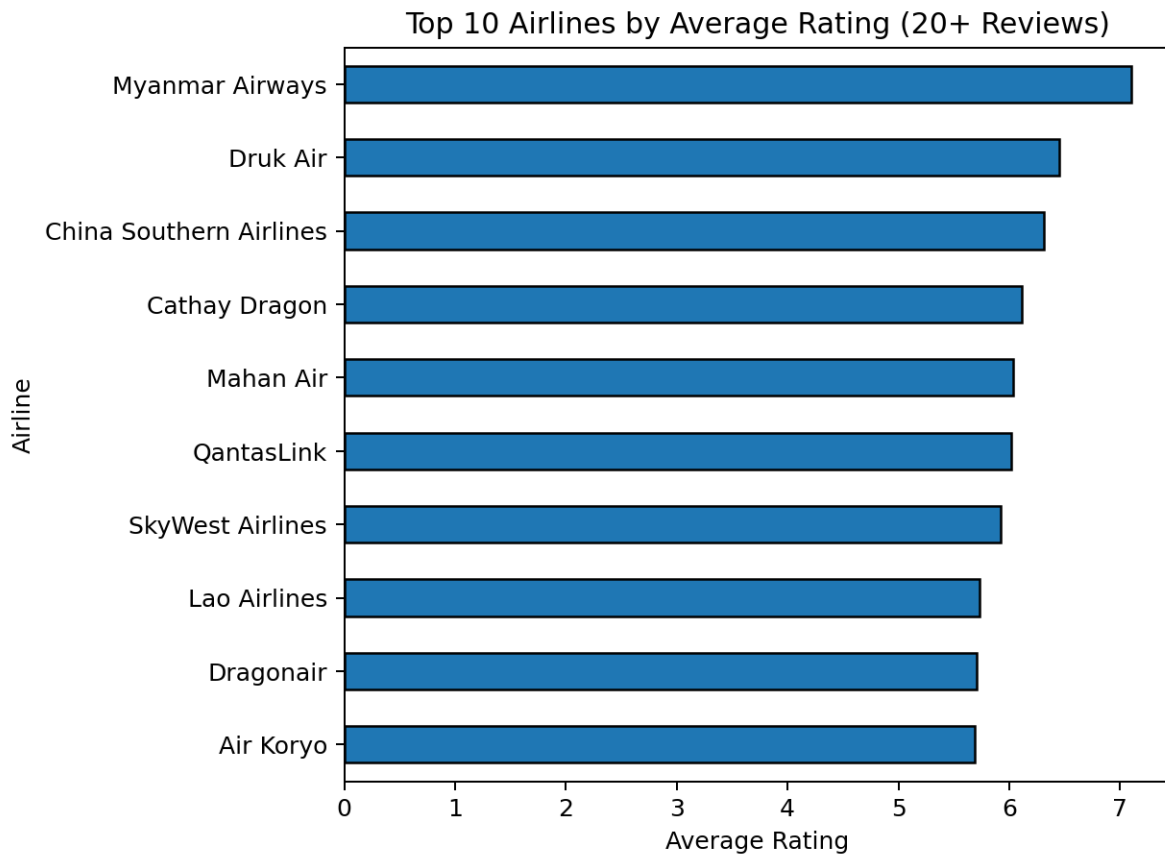


Figure 6. Top 10 airlines by average rating among airlines with at least 20 reviews.

Top airline	Reviews	Average rating	Recommendation rate
Myanmar Airways	20	7.10	95.0%
Druk Air	20	6.45	100.0%
China Southern Airlines	100	6.31	98.0%
Cathay Dragon	62	6.11	82.3%
Mahan Air	30	6.03	80.0%
QantasLink	99	6.02	80.8%
SkyWest Airlines	26	5.92	69.2%
Lao Airlines	72	5.74	90.3%
Dragonair	95	5.71	75.8%
Air Koryo	45	5.69	77.8%

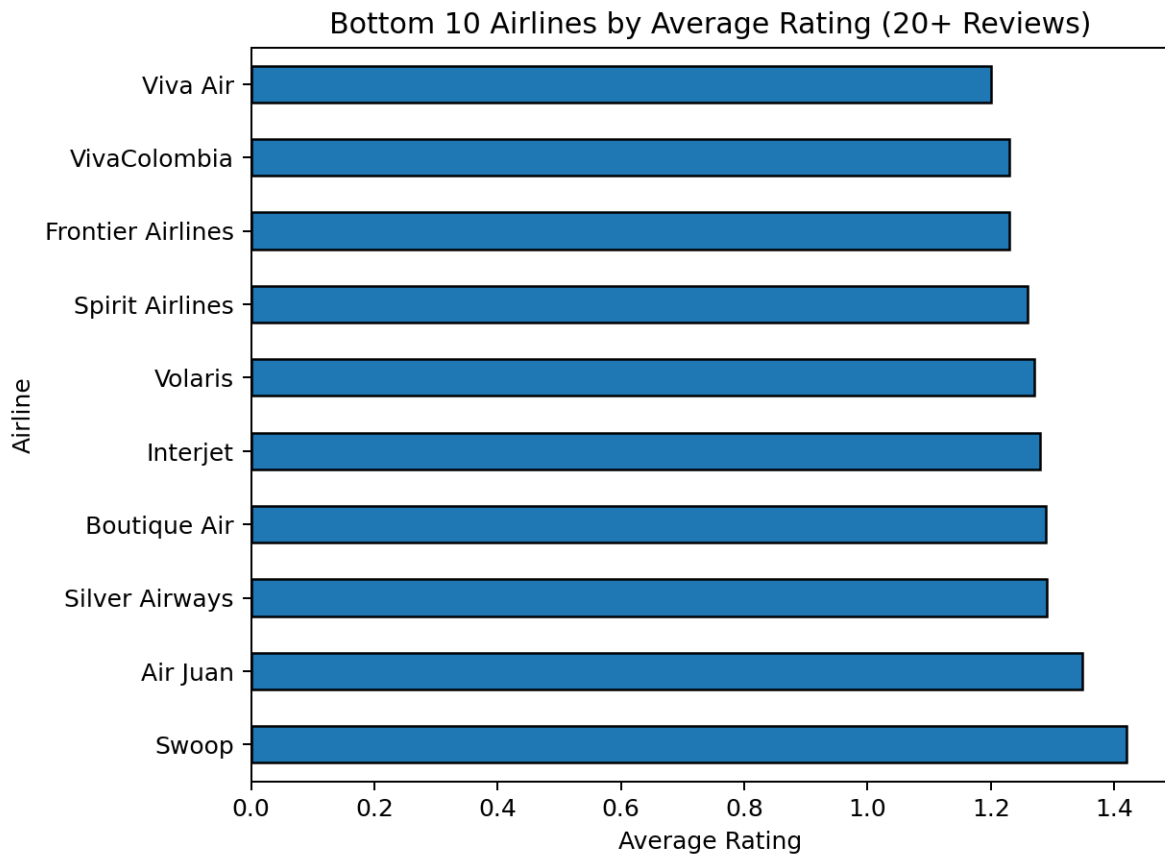


Figure 7. Bottom 10 airlines by average rating among airlines with at least 20 reviews.

Bottom airline	Reviews	Average rating	Recommendation rate
Viva Air	70	1.20	1.4%
VivaColombia	48	1.23	8.3%
Frontier Airlines	100	1.23	2.0%
Spirit Airlines	100	1.26	6.0%
Volaris	100	1.27	5.0%
Interjet	100	1.28	2.0%
Boutique Air	38	1.29	18.4%
Silver Airways	100	1.29	8.0%
Air Juan	23	1.35	4.3%
Swoop	100	1.42	10.0%

8. Statistical Analysis

The statistical analysis checks whether the observed patterns are likely to be meaningful rather than random noise. Since the final file has no date or text fields, the analysis focuses on rating-recommendation relationships and categorical associations.

Test / statistic	Result	Interpretation
Recommended vs not recommended rating difference	Recommended mean = 5.72; Not recommended mean = 1.76; Difference = 3.96	Passengers who recommend the airline give much higher ratings.
Welch t-test	$p < 0.001$	The rating difference between recommending and non-recommending

		passengers is statistically significant.
Effect size	Cohen's d = 1.82	The size of the difference is very large, not just statistically significant.
Rating vs recommendation correlation	r = 0.655	Overall rating has a strong positive relationship with recommendation behaviour.

Association tested	Statistical result	Business reading
Rating category	Chi-square = 10,737.6; df = 2; p < 0.001	Cramer's V = 0.693 (strong association)
Seat type	Chi-square = 642.7; df = 4; p < 0.001	Cramer's V = 0.170 (weak-to-moderate association)
Traveller type	Chi-square = 945.5; df = 4; p < 0.001	Cramer's V = 0.206 (moderate association)

Business interpretation: improving overall experience is likely to improve recommendation behaviour, but the project should still investigate service-level drivers in the raw dataset. Rating category is strongly linked with recommendation, while seat type and traveller type also show statistically significant differences. This supports a segmented action plan instead of one generic customer-service message for all passengers.

9. Tableau Dashboard Design

The Tableau dashboard is designed as a decision-support layer for customer experience analysis. The executive view should answer: How satisfied are passengers overall? Which segments are weak? Which airlines are outperforming or underperforming? The operational view should let users drill down into airline, seat type, traveller type, and recommendation status.

Dashboard element	Purpose	Recommended implementation
Executive KPI cards	Show total reviews, average rating, recommendation rate, low rating share, and number of airlines.	Place KPI cards across the top so the viewer understands the story within five seconds.
Rating distribution chart	Show whether ratings are concentrated at low, medium, or high values.	Use a histogram or bar chart grouped by rating_category.
Segment comparison	Compare satisfaction by seat_type and type_of_traveller.	Use horizontal bar charts so long category names are readable.
Airline benchmark view	Identify top and bottom airline performance.	Use average rating and recommendation rate together; apply a review-count filter.
Interactive filters	Allow decision-makers to narrow the analysis.	Filters: airline_name, seat_type, type_of_traveller, rating_category, is_recommended.

Dashboard URL: [Airline Analysis Dashboard - Customer Experience Overview](#)

Suggested dashboard layout

Area	Recommended content
Top band	Dashboard title and five KPI cards: Total Reviews, Airlines, Average Rating, Recommendation %, Low Rating Share.
Middle left	Rating distribution and rating category breakdown.
Middle right	Recommendation rate by rating category and seat type.
Bottom left	Average rating by traveller type.

Bottom right	Top/bottom airline benchmark table or bar chart.
Right side	Filters for airline, seat type, traveller type, rating category, and recommended flag.

10. Insights Summary

1. Customer dissatisfaction is the main business risk: 68.3% of reviews are in the Low category, so the project should focus first on recovery and root-cause analysis rather than only celebrating high performers.
2. The median rating is 1, meaning at least half of the reviewers gave a very low overall score. This signals a major experience gap in the review population.
3. Only 34.9% of passengers recommend the airline, so customer advocacy is weak and should be treated as a core KPI.
4. High-rated reviews convert into recommendations at 97.7%, while low-rated reviews convert at only 15.4%. Improving low ratings is likely to improve recommendation outcomes.
5. Business Class has the highest average rating among seat types at 4.33, but Economy Class has the largest volume with 18,853 reviews and a lower average rating of 3.00. This makes Economy the highest-priority commercial segment.
6. Premium Economy does not show a clear satisfaction premium in the final dataset; its average rating is 2.99, close to Economy. This could hurt customers who paid more but did not feel a stronger experience.
7. Family Leisure and Couple Leisure travellers are the weakest named segments, with average ratings of 2.56 and 2.74. These groups need more targeted service design.
8. The Unknown traveller category has high average scores but should not be interpreted as a real segment. It is a data-quality flag, not a customer persona.
9. Airline performance varies widely. Among airlines with 20+ reviews, the top average ratings are above 6, while the bottom group is close to 1.2-1.4. This gap is large enough to support benchmarking and recovery planning.
10. The correlation between overall rating and recommendation is 0.655, and the recommended vs non-recommended rating gap is 3.96 points. This makes recommendation behaviour a strong executive KPI for tracking customer experience improvement.

11. Recommendations

Priority	Insight	Recommendation	Expected business impact	Feasibility
1	Low ratings dominate and recommendation is weak.	Launch a low-rating recovery programme focused on the most common complaint drivers in Economy and Premium Economy.	Higher ratings, better recommendation rate, and reduced reputational risk.	Medium: requires service root-cause review, staff training, and tracking in Tableau.
2	Family and Couple Leisure travellers show the	Create a leisure-family service playbook covering	Improves satisfaction in high-emotion journeys	Medium: requires customer journey

	weakest named segment ratings.	seating together, baggage clarity, disruption communication, and airport support.	where bad experiences spread quickly through reviews.	mapping and coordination with airport operations.
3	Top and bottom airline ratings differ sharply.	Benchmark best-performing airlines and convert their strengths into a checklist for weaker airlines or service areas.	Turns comparison into a practical improvement roadmap instead of only ranking airlines.	Low-to-medium: mainly analytical work followed by operational adoption.
4	Current final CSV does not retain service-level driver columns.	In the next ETL version, keep seat comfort, cabin staff service, food, ground service, inflight entertainment, and value-for-money fields.	Enables true root-cause analysis and stronger recommendations.	Low: mostly a data pipeline and Tableau model update.
5	Incorrect KPI logic can mislead stakeholders, as seen in the 100% recommendation issue from bool conversion.	Create a KPI validation checklist before dashboard publication.	Improves trust in the dashboard and prevents wrong business conclusions.	Low: add automated checks in Python notebooks or scripts.

12. Impact Estimation

The dataset does not include revenue, booking value, or actual passenger counts, so the impact estimate is intentionally conservative and shown as an operational proxy rather than a financial claim. The logic is still useful: if the airline can move more passengers from low ratings to medium/high ratings, recommendation behaviour should improve.

Scenario	Assumption	Estimated impact on this review base
Conservative improvement	Recommendation rate improves by 5 percentage points.	About 1,116 additional recommending passengers out of 22,329 reviews.
Stronger improvement	Recommendation rate improves by 8 percentage points.	About 1,786 additional recommending passengers out of 22,329 reviews.
Low-rating reduction	Low rating share falls from 68.3% to about 60%.	Roughly 1,850 reviews would shift out of the Low category in a dataset of this size.
Dashboard governance	KPI validation prevents incorrect reporting.	Avoids wrong executive decisions, such as treating recommendation as 100% due to a coding error.

Why act now: the current satisfaction pattern is already heavily skewed toward low ratings. Waiting allows dissatisfaction to keep accumulating in public review channels. A dashboard-backed improvement cycle gives leadership a way to monitor whether interventions are actually changing passenger perception.

13. Limitations

- The final uploaded dataset has only six columns, so this report cannot calculate service-specific root causes from the final file.
- Review_date and date_flown are not present in the final file, so month-by-month trend analysis cannot be computed here.
- Review text is not present in the final file, so sentiment analysis and topic modelling are future-scope items.
- Public review data may be biased toward passengers with strong positive or negative experiences.

- Airline ratings are not weighted by actual passenger volume, route importance, ticket price, or operational disruptions.
- Unknown categories should be treated as data-quality indicators and not as meaningful customer groups.
- The report identifies associations, not causation. For example, Business Class has higher ratings, but the dataset alone does not prove the exact operational reason.

14. Future Scope

1. Retain all service-rating fields in the final Tableau dataset so the dashboard can explain why ratings are low.
2. Add review_date and date_flown to support time-series trends, seasonality checks, and pre/post intervention analysis.
3. Use review text for sentiment analysis, topic modelling, and complaint-driver extraction.
4. Build a predictive model to estimate recommendation probability from service scores, seat type, traveller type, and airline.
5. Add route and region-level analysis to identify whether dissatisfaction is concentrated in certain markets.
6. Create a scheduled ETL pipeline so the dashboard can refresh with new review data.
7. Add a data-quality dashboard showing missing fields, Unknown categories, and KPI validation checks.

15. Conclusion

SkyInsight turns airline review data into a practical customer experience decision tool. The analysis shows that passenger satisfaction is weak in the uploaded final dataset: the average rating is 3.14, the median is 1, 68.3% of reviews are Low, and only 34.9% of reviewers recommend the airline. The strongest business message is that improving low-rated experiences, especially in high-volume Economy and leisure segments, is likely to have the greatest effect on recommendation behaviour. The recommended next step is to use the Tableau dashboard as a monitoring layer while improving the ETL pipeline to keep service-level drivers for deeper root-cause analysis.

16. Appendix

A. Data dictionary

Field	Definition	Example
airline_name	Airline reviewed by the passenger.	AB Aviation
overall_rating	Overall satisfaction score from the review.	9.0
seat_type	Cabin or seat class.	Economy Class
type_of_traveller	Travel purpose or traveller group.	Solo Leisure
is_recommended	Binary recommendation flag: 1 = yes, 0 = no.	1
rating_category	Derived category from overall_rating: Low, Medium, or High.	High

B. Cleaning logic excerpt

```
df = pd.read_csv('../data/raw/airline_reviews.csv')
df.columns = df.columns.str.lower().str.strip().str.replace(' ', '_')
df['overall_rating'] = pd.to_numeric(df['overall_rating'], errors='coerce')
df['recommended'] = df['recommended'].str.lower().str.strip()
df['is_recommended'] = df['recommended'].map({'yes': 1, 'no': 0})
df = df.dropna(subset=['overall_rating'])
```

```
def rating_category(x):
    if x <= 3:
        return 'Low'
    elif x <= 6:
        return 'Medium'
    else:
        return 'High'
```

```
df['rating_category'] = df['overall_rating'].apply(rating_category)
df.to_csv('final_airline_reviews.csv', index=False)
```

C. References

- DVA Capstone 2 Project Report Outline: Uploaded PDF: [DVA_Capstone_2_Project_Report_Outline.pdf](#)
- Project GitHub Repository: https://github.com/yogeshm01/SectionD_G15_SkyInsight
- Kaggle Airline Reviews Dataset: <https://www.kaggle.com/datasets/juhibhojani/airline-reviews>
- Tableau Public Dashboard: <https://public.tableau.com/app/profile/bhavya.punj2072/viz/AirlineAnalysisDashboard1/CustomerExperienceOverview?publish=yes>
- Cleaned analysis file: final_airline_reviews(1).csv, uploaded to this ChatGPT conversation

17. Contribution Matrix

The contribution details below are aligned with the roles listed in the project repository. Final grading evidence should be checked against GitHub commits, pull requests, and submitted artifacts.

Team member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Yogesh Mishra	Primary	Primary	Support	Support	Support	Support	Support
Mohammad Affan Anas	Support	Support	Primary	Primary	Primary	Primary	Support
Aditi Singh	Support	Support	Support	Support	Support	Primary strategy inputs	Support

Bhavya Punj	Support	Support	Support	Support	Primary dashboard publication / quality support	Quality review	Primary
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Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: _____ Date: _____